

**MI&CT ESERVICE NOTE
SN07051510b**

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This short-term informational note is invalid 6 months after the issue date.

Date: July 2, 2007

Subject: Customer Password Management Process

Effectivity:
All Lightspeed CT Scanners and PET Scanner running the LINUX Operating System.

Details:
More and more customers are requesting that GE change passwords for UNIX accounts on our CT and PET systems. They want either to modify the default passwords on their own or have GE Field Personnel change them periodically as part of a standard routine. Presently, we do not have a procedure to do this and “Ad hoc” solutions are invented which vary from customer to customer.

Resolution:
This Service Note describes the process of how to manage the passwords to improve the security level on networked systems. It also contains information for generating more robust passwords to reduce the security risk which simple passwords present.

Process Overview

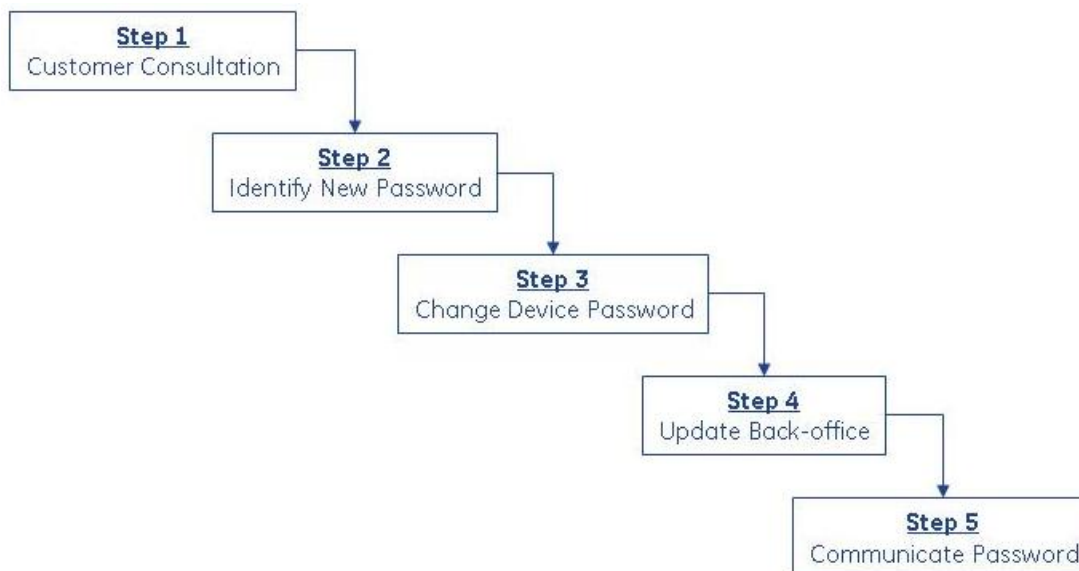


Illustration 1

Step 1 – Customer Consultation

When the customer or GE initiates this process, the GE FE is to coordinate a customer consultation to discuss password provisions. As a result of that meeting the following process must be followed to complete the password change accurately.

Note: Many systems have the ability to enable "NIS", a feature that uses a central server to authenticate logins. This means passwords for some accounts are managed by the customer in their database and not on the product itself. If NIS is enable then some passwords may still get changed, however this will require additional technical conversations with the customer so online engineers should get included in these situations.

Step 2 – Identify New Password(s)

We recommend changing passwords on all accounts used during remote connectivity:

- o Root
- o Ctuser
- o Insite
- o Service

The customer may request specific passwords for root and the user account (ctuser), but only GE service should know the others.

To create new passwords, use the following ways:

- If required, use customer rules and guidelines for password creation.
- If the customer does not have a defined set of rules or guidelines, use the following guidelines to develop a strong password:
 - o Must be 8 characters min.
 - o Cannot be blank or left as the default.
 - o Should contain a mix of numbers, alpha, and special characters.
 - o Must not be made up solely of dictionary words.
 - o May contain the system ID with at least 1 more character.
 - o Should not be the same value at different sites.

Each account on a single system should have a unique password. Eg. the "root" and "insite" passwords should be different from each other. For convenience, the same password value for a single account can be used on multiple systems at a site or customer. Eg. the "insite" user may have the same password value on several different systems in a hospital. However, don't use the same value over multiple sites or across a region, as that mimics the original default value problem.

Good password examples:

!414555MR5
4\$42CTAW
#big996622LS16
2getgre0552

Bad password examples:

414555MR5
operator
123456789a

Step 3 – Change Device Password(s)

After the passwords are identified, the FE should make the password changes on the device following detailed instructions below:

Note: If you make a typing error, type <ctrl-c><Enter> to exit and start over. This is important because Linux will include the typing correction in the password, making it unusable. For similar reasons do not use copy/paste to set a new password, always type it twice. If the root password gets set with a bad value a LFC may be needed to recover.

1. For root:

- a. Click *Unix Shell* in the upper right of the monitor and put the cursor in the shell window. Enter the following in the telnet window:
- b. Type: **su -** Press [Enter]
- c. Type the current password Press [Enter]
- d. Type: **passwd** Press [Enter]
- e. Type the new password Press [Enter]
- f. Retype the new password Press [Enter]
- g. Verify message: Passwd: all authentication tokens updated sucessfully
- h. Type: **exit** Press [Enter]
- i. Type: **exit** Press [Enter]

2. For ctuser:

- a. Click *Unix Shell* in the upper right of the monitor and put the cursor in the shell window. Enter the following in the telnet window:
- b. Type: **passwd** Press [Enter]
- c. Type the current password Press [Enter]
- d. Type the new password Press [Enter]
- e. Retype the new password Press [Enter]
- f. Verify message: Passwd: all authentication tokens updated sucessfully
- g. Type: **exit** Press [Enter]

3. For insite:

- a. Click *Unix Shell* in the upper right of the monitor and put the cursor in the shell window. Enter the following in the telnet window:
- b. Type: **su insite** Press [Enter]
- c. Type the current password Press [Enter]
- d. Type: **passwd** Press [Enter]
- e. Type the current password Press [Enter]
- f. Type the new password Press [Enter]
- g. Retype the new password Press [Enter]
- h. Verify message: Passwd: all authentication tokens updated sucessfully
- i. Type: **exit** Press [Enter]
- j. Type: **exit** Press [Enter]

4. For service:

- a. Click *Unix Shell* in the upper right of the monitor and put the cursor in the shell window. Enter the following in the telnet window:
- b. Type: **su service** Press [Enter]
- c. Type the current password Press [Enter]
- d. Type: **passwd** Press [Enter]
- e. Type the current password Press [Enter]

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- f. Type the new password Press [Enter]
- g. Retype the new password Press [Enter]
- h. Verify message: Passwd: all authentication tokens updated sucessfully
- i. Type: **exit** Press [Enter]
- j. Type: **exit** Press [Enter]

Note: The following errors may occur, preventing a password (including the defaults) being set from within a user account:

BAD PASSWORD: it is based on a dictionary word

BAD PASSWORD: it is too similar to the old one

BAD PASSWORD: it is too simplistic/systematic

In these cases it is best to choose a stronger password, but a work-around is to become **root** and use the command **passwd <account>** to set the selected password, over-riding the warnings.

Step 4 – Update Password(s) in Connectivity Database

To checkout the system password follow the following procedure:

1. Contact the checkout team at 866-759-7075.
2. Inform the technician you are making password changes. Provide the System ID, and ask them to do a checkout with all the new password values.
3. The checkout technician will verify the system ID and new password values. To reduce miscommunication of the password verbally, consider using the phonetic alphabet listed below.

A – Alpha	B – Bravo	C – Charlie	D – Delta
E – Echo	F – Foxtrot	G – Golf	H – Hotel
I – India	J – Juliet	K – Kilo	L – Lima
M – Mike	N – November	O – Oscar	P – Peter
Q – Quebec	R – Robert	S – Sierra	T – Tango
U – Umbrella	V – Victor	W – Whiskey	X – X-Ray
Y – Yankee	Z – Zulu		

4. The technician will run a checkout and password verification procedure to update the password(s) and inform you when the process is complete.

Step 5 – Communicate New Password(s)

After completing the checkout, follow your customer's guidelines for password communication and storage. Inform the customer of the new passwords with the exception of those used for remote service only (e.g. insite and service).

If the customer approves, write down the new passwords and store them in a secure location on site. The service note **Appendix A** includes a sample form to place in a logbook or tape inside a cabinet.

In the situation where a customer wants to know more about what GE does with passwords, escalate to the service security team

(http://supportcentral.ge.com/products/sup_products.asp?prod_id=24038).

Deviations

In the case where a customer directly calls the OLC requesting a password update, the online engineer should follow the same steps outlined in this document. Some online engineers have

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access to make password changes in the back office databases, so they may execute Step 4 themselves without calling the checkout team.

Important – Make sure to email a copy of the case to the site primary field engineer to complete Step 5. This will ensure that local service has documentation of the change to access the system.

If you experience any problems with this procedure please contact field leadership or the Online Center.

Affected Service Publication: Linux Based CT/PET systems

Document revision Changes:

Rev.B

Step 1: Added a note about NIS.

Step 2: Specified accounts to change, added note to clarify duplication.

Step 3: Corrected procedure. Added a method for “difficult” passwords.

Scott B. Killen - CISSP
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Appendix A



Password Record for
System ID _____
By _____
Date _____

Login Id	
Password	

Login Id	
Password	

Login Id	
Password	

Login Id	
Password	

Login Id	
Password	